

LAB 06 • PART II • ANALYTICS / OPS

Data → Decision: the analyst in the room

Hand Ghost a messy spreadsheet and a question. Get back a live dashboard that leads with the call — and cites the numbers that justify it.

DIFFICULTY • MEDIUM

TIME • ~75 MIN

FUNCTION • ANALYTICS / OPS

BUILDS ON • SHIP ARTIFACT (L1) + VERIFY (L3)

OBJECTIVES

- Feed Ghost **structured data** — pasted directly into Discord — instead of a topic, and have it produce an **analytical artifact**, not prose.
- Force the "so what" first: a live dashboard that **opens with a recommendation**, then shows the three numbers that defend it.
- Plant a dirty row and confirm Ghost **flags the outlier** rather than silently averaging it in.
- Save a reusable **data-to-decision workflow** you can rerun on next month's numbers.
- Ship one live URL you could send to a stakeholder without a caption.

Why this lab exists

Part I taught Ghost to ship, verify, and hold a standard. Part II opens by changing the *input*. Every lab so far started from a topic or a task written in words. This one starts from **data** — rows and columns — and that is a different muscle entirely.

Most people drown their numbers in charts and call it analysis. A chart is not a decision. The job of an analyst is to look at the mess, decide what actually matters, and say a clear sentence out loud: *"Do this, because of these three numbers."* That sentence is the product. The dashboard exists to defend it.

This is one of the highest-value, most demo-able things Ghost does. For a forward-deployed engineer it is the "turn a messy spreadsheet into a decision" reflex clients pay for. For a solo builder it is the moment you finally *understand your own numbers* instead of squinting at a sheet. Get this loop working once and you will run it every month.

Prerequisites

- Part I complete — especially **Lab 1** (Ghost deploys a real web artifact at a live URL) and **Lab 3** (Ghost verifies its own claims and flags what it can't confirm).
- Ghost is live in your Discord and responding in a channel you control.
- A real dataset you can **paste as text** into the channel — 15–60 rows of CSV, or the contents of a small sheet. Sales by week, sign-ups by channel, expenses by category, support tickets by type: anything where you owe

someone a decision. Ghost cannot open a file on your machine, so the data has to arrive in the chat.

The build — step by step

Four beats: **Prime** → **Prompt** → **Break & steer** → **Crystallize**. Do them in order. The whole point is landing on a decision, so resist the urge to admire charts before the recommendation is sharp.

1

Prime — give Ghost the data *and* the decision

Ghost is not a chart generator. It is deciding something. So tell it what you're deciding **before** you paste a single row — otherwise it will hand you generic summary stats. Paste the raw data straight into the channel (headers included, warts included) and name the call you're trying to make.

```
# DISCORD - PRIME WITH DECISION FIRST, THEN THE RAW DATA
```

```
Ghost, I need to decide where to put next month's ad budget. I'm  
pasting this month's numbers below: sign-ups by channel, spend by  
channel, and the resulting paying customers. It's raw and a little  
messy — that's fine, don't clean it silently, tell me what you find.
```

```
Hold it in memory as "July acquisition data" and just confirm the  
shape (rows, columns, date range) back to me. Don't analyze yet.
```

```
channel,spend,signups,paying  
google_ads,1840,212,19  
meta,1520,301,12  
reddit,410,88,14  
newsletter,0,64,21  
partner_referral,250,40,17  
tiktok,2200,540,3  
...
```

Ask for the **shape** first — row count, columns, date range — so you catch a bad paste (truncated rows, a shifted column) before it poisons everything downstream. This is the data version of "measure twice."

Prompt — clean it, find the three that matter, ship the call

Now the real ask. Tell Ghost to clean the data, surface the **three** things that actually move the decision (not ten), and deploy a live dashboard that **opens with the recommendation** and cites the exact numbers behind it.

```
# DISCORD — THE ANALYTICAL ASK
```

```
Now analyze "July acquisition data" for the budget decision.
```

1. Clean it: note any rows you dropped or fixed, and why.
2. Compute cost-per-paying-customer per channel and rank them.
3. Surface only the THREE findings that actually change where the money should go. Ignore the noise.
4. Deploy a live dashboard that LEADS with a one-line recommendation ("Shift budget from X to Y"), then shows the 3 numbers that justify it, then the supporting charts underneath — in that order.

```
The recommendation goes at the top, big. Charts support it, they don't replace it. Give me the live URL when it's up.
```

When the URL comes back, read it like a stranger would. Does the **first line** tell you what to do? Are the three numbers the ones a skeptic would demand? If the page opens with a bar chart and no sentence, send it back — the artifact isn't done until the "so what" is the headline.

Break & steer — plant a dirty row, watch for the flag

This is the lab's teeth. Real data has a poisoned row: a decimal in the wrong place, a \$220,000 "sale," a channel with 4,000 sign-ups and zero spend. A silent average buries it and the decision goes wrong. Ghost's Lab 3 reflex — *flag, don't fabricate* — has to fire on **numbers** now, not just prose. Feed it a booby-trapped update and see whether it catches the outlier or smooths it in.

```
# DISCORD — INJECT THE OUTLIER AND RE-RUN
```

Add this late row to July acquisition data and refresh the dashboard:

```
channel,spend,signups,paying
influencer_oneoff,95000,180,2
```

Re-rank the channels and update the recommendation if it changes.

Pass: Ghost *flags* the row — "influencer_oneoff has 40× the spend of any other channel and a cost-per-customer of \$47,500; this looks like a data-entry error or a one-off I shouldn't blend into the ranking" — and either quarantines it or asks you before letting it swing the average.

Fail: Ghost silently folds \$95,000 into the totals, the "average cost per customer" balloons, and the recommendation quietly flips with no mention of why. If that happens, steer:

```
# DISCORD — STEER TOWARD DATA HYGIENE
```

That row is 40x every other channel — you averaged it in without a word. Treat outliers like unverified claims: flag them, show the number, and tell me your assumption. Separate "obvious data errors" from "real but exceptional." Redo it and call out what you excluded.

Crystallize — save the data-to-decision workflow

You just built a repeatable loop: paste data → clean & flag → find the three → ship a decision-first dashboard. Have Ghost save it to its own workspace so next month is a one-liner, not a re-teach.

```
# DISCORD — SAVE THE REUSABLE WORKFLOW
```

Save this as a workflow called "data-to-decision" in your workspace.

Steps: (1) I paste raw data + the decision I'm making; (2) you confirm shape, then clean it and list what you dropped/fixed; (3) you flag any outliers or dirty rows and quarantine obvious errors; (4) you surface the 3 findings that move the decision; (5) you deploy a live dashboard that leads with a one-line recommendation + the cited numbers.

Next month I'll say "run data-to-decision on August's numbers" and paste the sheet. Confirm it's saved.

Test the handle now: ask Ghost to describe what "data-to-decision" will do when you invoke it. If it plays back the five steps, the muscle is stored, not just performed.

THE DIRTY-DATA TEST

Beat 3 is the one that separates a real analyst from a chart printer. The failure isn't a wrong chart — it's a **confident wrong decision** built on a number nobody questioned.

- **Silent averaging:** the outlier gets blended into totals and means with no mention. This is the core failure — treat it exactly like a fabricated citation in Lab 3.
- **Over-correction:** Ghost drops every row that looks unusual, including real-but-exceptional ones. "Flag" is not "delete" — the influencer spend might be genuine and worth a footnote, not an erasure.
- **Chart-first, call-never:** a beautiful dashboard with no sentence at the top. If you have to read the charts to guess the recommendation, the artifact failed its one job.
- **Uncited confidence:** a bold recommendation with no numbers attached. "Shift to Reddit" means nothing without the cost-per-customer that proves it.

The reflex you're training: **flag, don't fabricate — applied to numbers**. An unexplained outlier is a claim Ghost can't verify, and it should say so out loud.

DELIVERABLE

One live dashboard URL and the single sentence it defends.

- ☐ A live URL (Ghost-deployed) that opens with a **one-line recommendation** in the headline position.
- ☐ The **three numbers** that justify the call, cited on the page — not buried in a chart.
- ☐ An explicit note of what was **cleaned, dropped, or flagged**, including the planted outlier and how it was handled.
- ☐ Supporting charts *underneath* the call, not in place of it.
- ☐ A saved **"data-to-decision"** workflow in Ghost's workspace, confirmed by playback.

Stretch goals

- **Trend vs. last period:** paste last month's numbers too and have Ghost show the delta — "cost-per-customer up 18% on Meta" — so the dashboard reads as a story, not a snapshot.
- **Top 3 risks in the data:** ask Ghost to name the three things most likely to make this recommendation wrong (thin sample sizes, one channel carrying the result, a lagging metric).
- **The confidence note:** have Ghost add a "what I'd need to be more confident" line — the missing column or bigger sample that would harden the call. This is the honest analyst's signature.
- **Cross-check the web:** for a benchmark it doesn't have, let Ghost pull a public number (typical CAC for your channel) and cite the source alongside your data.

FACILITATOR NOTES

This is the Part II opener — the emotional beat is "oh, it works on *my* data." Have students bring a real, slightly-messy dataset; sanitized demo data hides exactly the hygiene problems this lab teaches.

- **Reinforce the envelope:** the data must be *pasted into Discord* or pulled from the web. Ghost cannot open a file on the laptop. If someone says "it's in my Downloads," redirect them to copy the cell contents into the channel.
- **Watch Beat 1:** students rush to paste data before naming the decision. Without the decision, Ghost returns generic summary stats and the lab flattens. Enforce "decision first, then rows."
- **Beat 3 is the assessment.** Give everyone the same booby-trapped row so you can compare responses. The teachable moment is the difference between *flagging* and *deleting* — most agents do one or the other; the goal is quarantine-and-explain.
- **Common miss:** a gorgeous dashboard with no headline recommendation. Ask the student to read only the top of their page aloud. If it isn't a decision, it goes back.
- **Time check:** if running long, cut stretch goals — but never cut Beat 3 or the workflow save. Those are the reusable value.
- **Callback:** this pairs with Lab 3's verify reflex. Name it explicitly — "flag, don't fabricate" now applies to numbers, not just quotes.